Abdelrahman Elawady

Madinah, Saudi Arabia | +966544776316 | ab.elawady@gmail.com | www.linkedin.com/in/abdelrahmanelawady

Electrical Engineering student combining expertise in industrial automation and control systems with award-winning academic excellence, competitive hardware prototyping and advanced scientific research in top journals.

Education

University of Prince Muqrin UPM, Madinah, Saudi Arabia — Sep 2023 – Expected 2027

- B.Sc. in Electrical Engineering – GPA: 3.99/4
- Relevant Courses: Programmable logic controllers (PLC), Control systems analysis, Introduction to Artificial Intelligence – IBM

Experience

Professional Summer training intern

Jabal Al-Alam company JAA (Delta Controls partner) – Madinah, Saudi Arabia — *Jun 2025 – Aug 2025*

- Assembled and configured 500+ GRMS classic control electrical panels with relays and controllers.
- Drafted I/O lists in Excel for GRMS and BMS applications, ensuring accurate mapping and efficient system configuration.
- Programmed building automation controllers using GCL+, and configured database software objects via BACnet, MS/TP/Ethernet/IP and Modbus network protocols.
- Assisted in testing, commissioning, and troubleshooting of BMS field devices and panels (MCCs, DDCs, AHUs, Chillers, pumps), and created AutoCAD layouts, Nameplates laser design and real-time monitoring graphics.

Academic projects

Automated Hospital Logistics PLC Conveyor System — *Ongoing*

- Designing ladder logic control structures to manage automated material handling and bin staging for a hospital logistics conveyor.
- Developing a predictive supply and demand forecasting AI regression model techniques to optimize routing efficiency.

*Ionic Plasma Thruster project – **1st place in EE All-levels competition*** — *May 2025*

- Engineered a fuel-less Ionic propulsion system using high-voltage corona discharge with cost-effective materials.
- Designed and implemented a ZVS flyback soft-switching driver circuit with voltage multiplier, achieving efficient high-voltage AC generation and enabling a fully functional, 3D-printed boat prototype to solve a sailing problem with space technology.

Smart LED Display Systems for Campus Offices — *Nov 2024 – Feb 2025*

- Pioneered and designed Bluetooth-enabled LED display systems for the Dean's Office and Career Center using ESP32 microcontrollers, optimizing RAM and DMA allocation to ensure flicker-free rendering on a low-latency UART interface.
- Deployed custom web app and engineered cost-effective solutions that reduced implementation costs by approximately 75% compared to commercial alternatives with low maintenance for professional environments.

*Electric generator project – **2nd place in Physics competition*** — *May 2024*

- Collaborated with a colleague in a trial-and-error approach to design and construct a low-budget DIY axial flux electric generator using 250-loop coils and magnets mounted on a CD disk, resulting in 3V output.

Publications

- **Elawady, A.**, Mohamed, J. A., & Bin Abid, M. (2025). Advancements in microplastics detection techniques and their multidimensional impacts on aquatic ecosystems and human health. *Journal of Hazardous Materials: Plastics* (**Q1**).
- **Elawady, A.,** et al. (2025). Polymer-Based Innovations in Concrete: A Review on Enhancing Self-Healing Capabilities. *Innovative Infrastructure Solutions* (**Q2**).
- Bin Abid, M., **Elawady, A.**, & Shamim, R. (2025). Exploring AlUla, Saudi Arabia: a journey through history, heritage, resources, and tourism. *GeoJournal* (**Q2**).

Personal solo projects

Personal Portfolio website — *Dec 2025*

- Programmed my own dynamic engineering portfolio website: https://abdelrahman-elawady.github.io/portfolio/

Python Interactive covid-19 data visualization map — *Jun 2021*

- Developed an online interactive Covid-19 data map website using python folium library and HTML I-frames.

Skills & Technologies

Engineering Tools: AutoCAD, TIA portal, MATLAB, Simulink, Multisim, KiCad, MS office (Word, Excel, PowerPoint).
Software/ Programming: PLC Ladder Logic, Assembly, C++, Python, AI Pytorch, MATLAB, HTML, CSS, JavaScript, Git.
Hardware: ESP32, Arduino, PCB design, soldering, Electronics & IoT, 3D printing, Oscilloscope.

Leadership & Awards

- Electrical Engineering club leader, UPM – Feb 2025 – Sep 2025
- Prince Faisal bin Salman Award for academic excellence, UPM – 2025